Springer Texts in Statistics

Series Editors

G. Allen, Department of Statistics, Rice University, Houston, TX, USA

R. De Veaux, Department of Mathematics and Statistics, Williams College, Williamstown, MA, USA

R. Nugent, Department of Statistics, Carnegie Mellon University, Pittsburgh, PA, USA

Springer Texts in Statistics (STS) includes advanced textbooks from 3rd- to 4th-year undergraduate levels to 1st- to 2nd-year graduate levels. Exercise sets should be included. The series editors are currently Genevera I. Allen, Richard D. De Veaux, and Rebecca Nugent. Stephen Fienberg, George Casella, and Ingram Olkin were editors of the series for many years.

Gareth James • Daniela Witten • Trevor Hastie
Robert Tibshirani • Jonathan Taylor

An Introduction to Statistical Learning

with Applications in Python

Springer

Gareth James
Goizueta Business School
Emory University
Atlanta, GA, USA

Trevor Hastie
Departments of Statistics
and Biomedical Data Science
Stanford University
Stanford, CA, USA

Jonathan Taylor
Department of Statistics
Stanford University
Stanford, CA, USA

Daniela Witten
Departments of Statistics
and Biostatistics
University of Washington
Seattle, WA, USA

Robert Tibshirani
Departments of Biomedical
Data Science and Statistics
Stanford University
Stanford, CA, USA

ISSN 1431-875X ISSN 2197-4136 (electronic)
Springer Texts in Statistics
ISBN 978-3-031-38746-3 ISBN 978-3-031-38747-0 (eBook)
https://doi.org/10.1007/978-3-031-38747-0

© Springer Nature Switzerland AG 2023
This work is subject to copyright. All rights are reserved by the Publisher, whether the whole or part of the material is concerned, specifically the rights of translation, reprinting, reuse of illustrations, recitation, broadcasting, reproduction on microfilms or in any other physical way, and transmission or information storage and retrieval, electronic adaptation, computer software, or by similar or dissimilar methodology now known or hereafter developed.
The use of general descriptive names, registered names, trademarks, service marks, etc. in this publication does not imply, even in the absence of a specific statement, that such names are exempt from the relevant protective laws and regulations and therefore free for general use.
The publisher, the authors, and the editors are safe to assume that the advice and information in this book are believed to be true and accurate at the date of publication. Neither the publisher nor the authors or the editors give a warranty, expressed or implied, with respect to the material contained herein or for any errors or omissions that may have been made. The publisher remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

This Springer imprint is published by the registered company Springer Nature Switzerland AG
The registered company address is: Gewerbestrasse 11, 6330 Cham, Switzerland

To our parents:

Alison and Michael James

Chiara Nappi and Edward Witten

Valerie and Patrick Hastie

Vera and Sami Tibshirani

John and Brenda Taylor

and to our families:

Michael, Daniel, and Catherine

Tessa, Theo, Otto, and Ari

Samantha, Timothy, and Lynda

Charlie, Ryan, Julie, and Cheryl

Lee-Ann and Isobel

Preface

Statistical learning refers to a set of tools for *making sense of complex datasets*. In recent years, we have seen a staggering increase in the scale and scope of data collection across virtually all areas of science and industry. As a result, statistical learning has become a critical toolkit for anyone who wishes to understand data — and as more and more of today's jobs involve data, this means that statistical learning is fast becoming a critical toolkit for *everyone*.

One of the first books on statistical learning — *The Elements of Statistical Learning* (ESL, by Hastie, Tibshirani, and Friedman) — was published in 2001, with a second edition in 2009. ESL has become a popular text not only in statistics but also in related fields. One of the reasons for ESL's popularity is its relatively accessible style. But ESL is best-suited for individuals with advanced training in the mathematical sciences.

An Introduction to Statistical Learning, With Applications in R (ISLR) — first published in 2013, with a second edition in 2021 — arose from the clear need for a broader and less technical treatment of the key topics in statistical learning. In addition to a review of linear regression, ISLR covers many of today's most important statistical and machine learning approaches, including resampling, sparse methods for classification and regression, generalized additive models, tree-based methods, support vector machines, deep learning, survival analysis, clustering, and multiple testing.

Since it was published in 2013, ISLR has become a mainstay of undergraduate and graduate classrooms worldwide, as well as an important reference book for data scientists. One of the keys to its success has been that, beginning with Chapter 2, each chapter contains an `R` lab illustrating how to implement the statistical learning methods seen in that chapter, providing the reader with valuable hands-on experience.

However, in recent years `Python` has become an increasingly popular language for data science, and there has been increasing demand for a `Python`-

based alternative to ISLR. Hence, this book, *An Introduction to Statistical Learning, With Applications in Python* (ISLP), covers the same materials as ISLR but with labs implemented in `Python` — a feat accomplished by the addition of a new co-author, Jonathan Taylor. Several of the labs make use of the `ISLP Python` package, which we have written to facilitate carrying out the statistical learning methods covered in each chapter in `Python`. These labs will be useful both for `Python` novices, as well as experienced users.

The intention behind ISLP (and ISLR) is to concentrate more on the applications of the methods and less on the mathematical details, so it is appropriate for advanced undergraduates or master's students in statistics or related quantitative fields, or for individuals in other disciplines who wish to use statistical learning tools to analyze their data. It can be used as a textbook for a course spanning two semesters.

We are grateful to these readers for providing valuable comments on the first edition of ISLR: Pallavi Basu, Alexandra Chouldechova, Patrick Danaher, Will Fithian, Luella Fu, Sam Gross, Max Grazier G'Sell, Courtney Paulson, Xinghao Qiao, Elisa Sheng, Noah Simon, Kean Ming Tan, Xin Lu Tan. We thank these readers for helpful input on the second edition of ISLR: Alan Agresti, Iain Carmichael, Yiqun Chen, Erin Craig, Daisy Ding, Lucy Gao, Ismael Lemhadri, Bryan Martin, Anna Neufeld, Geoff Tims, Carsten Voelkmann, Steve Yadlowsky, and James Zou. We are immensely grateful to Balasubramanian "Naras" Narasimhan for his assistance on both ISLR and ISLP.

It has been an honor and a privilege for us to see the considerable impact that ISLR has had on the way in which statistical learning is practiced, both in and out of the academic setting. We hope that this new `Python` edition will continue to give today's and tomorrow's applied statisticians and data scientists the tools they need for success in a data-driven world.

It's tough to make predictions, especially about the future.

-Yogi Berra

Contents

Preface **vii**

1 Introduction **1**

2 Statistical Learning **15**
2.1 What Is Statistical Learning? 15
2.1.1 Why Estimate f? 17
2.1.2 How Do We Estimate f? 20
2.1.3 The Trade-Off Between Prediction Accuracy and Model Interpretability 23
2.1.4 Supervised Versus Unsupervised Learning 25
2.1.5 Regression Versus Classification Problems 27
2.2 Assessing Model Accuracy 27
2.2.1 Measuring the Quality of Fit 28
2.2.2 The Bias-Variance Trade-Off 31
2.2.3 The Classification Setting 34
2.3 Lab: Introduction to Python 40
2.3.1 Getting Started 40
2.3.2 Basic Commands 40
2.3.3 Introduction to Numerical Python 42
2.3.4 Graphics 48
2.3.5 Sequences and Slice Notation 51
2.3.6 Indexing Data 51
2.3.7 Loading Data 55
2.3.8 For Loops 59
2.3.9 Additional Graphical and Numerical Summaries . . 61
2.4 Exercises 63

3 Linear Regression **69**
3.1 Simple Linear Regression 70
3.1.1 Estimating the Coefficients 71
3.1.2 Assessing the Accuracy of the Coefficient Estimates 72
3.1.3 Assessing the Accuracy of the Model 77
3.2 Multiple Linear Regression 80
3.2.1 Estimating the Regression Coefficients 81

3.2.2 Some Important Questions 83
3.3 Other Considerations in the Regression Model 91
3.3.1 Qualitative Predictors 91
3.3.2 Extensions of the Linear Model 94
3.3.3 Potential Problems 100
3.4 The Marketing Plan 109
3.5 Comparison of Linear Regression with K-Nearest Neighbors 111
3.6 Lab: Linear Regression 116
3.6.1 Importing packages 116
3.6.2 Simple Linear Regression 117
3.6.3 Multiple Linear Regression 122
3.6.4 Multivariate Goodness of Fit 123
3.6.5 Interaction Terms 124
3.6.6 Non-linear Transformations of the Predictors . . . 125
3.6.7 Qualitative Predictors 126
3.7 Exercises 127

4 Classification 135
4.1 An Overview of Classification 135
4.2 Why Not Linear Regression? 136
4.3 Logistic Regression 138
4.3.1 The Logistic Model 139
4.3.2 Estimating the Regression Coefficients 140
4.3.3 Making Predictions 141
4.3.4 Multiple Logistic Regression 142
4.3.5 Multinomial Logistic Regression 144
4.4 Generative Models for Classification 146
4.4.1 Linear Discriminant Analysis for $p = 1$ 147
4.4.2 Linear Discriminant Analysis for $p > 1$ 150
4.4.3 Quadratic Discriminant Analysis 156
4.4.4 Naive Bayes 158
4.5 A Comparison of Classification Methods 161
4.5.1 An Analytical Comparison 161
4.5.2 An Empirical Comparison 164
4.6 Generalized Linear Models 167
4.6.1 Linear Regression on the Bikeshare Data 167
4.6.2 Poisson Regression on the Bikeshare Data 169
4.6.3 Generalized Linear Models in Greater Generality . 172
4.7 Lab: Logistic Regression, LDA, QDA, and KNN 173
4.7.1 The Stock Market Data 173
4.7.2 Logistic Regression 174
4.7.3 Linear Discriminant Analysis 179
4.7.4 Quadratic Discriminant Analysis 181
4.7.5 Naive Bayes 182
4.7.6 K-Nearest Neighbors 183
4.7.7 Linear and Poisson Regression on the Bikeshare Data 188
4.8 Exercises 193

5 Resampling Methods 201
5.1 Cross-Validation . . . 202
5.1.1 The Validation Set Approach . . . 202
5.1.2 Leave-One-Out Cross-Validation . . . 204
5.1.3 k-Fold Cross-Validation . . . 206
5.1.4 Bias-Variance Trade-Off for k-Fold Cross-Validation . . . 208
5.1.5 Cross-Validation on Classification Problems . . . 209
5.2 The Bootstrap . . . 212
5.3 Lab: Cross-Validation and the Bootstrap . . . 215
5.3.1 The Validation Set Approach . . . 216
5.3.2 Cross-Validation . . . 217
5.3.3 The Bootstrap . . . 220
5.4 Exercises . . . 224

6 Linear Model Selection and Regularization 229
6.1 Subset Selection . . . 231
6.1.1 Best Subset Selection . . . 231
6.1.2 Stepwise Selection . . . 233
6.1.3 Choosing the Optimal Model . . . 235
6.2 Shrinkage Methods . . . 240
6.2.1 Ridge Regression . . . 240
6.2.2 The Lasso . . . 244
6.2.3 Selecting the Tuning Parameter . . . 252
6.3 Dimension Reduction Methods . . . 253
6.3.1 Principal Components Regression . . . 254
6.3.2 Partial Least Squares . . . 260
6.4 Considerations in High Dimensions . . . 262
6.4.1 High-Dimensional Data . . . 262
6.4.2 What Goes Wrong in High Dimensions? . . . 263
6.4.3 Regression in High Dimensions . . . 265
6.4.4 Interpreting Results in High Dimensions . . . 266
6.5 Lab: Linear Models and Regularization Methods . . . 267
6.5.1 Subset Selection Methods . . . 268
6.5.2 Ridge Regression and the Lasso . . . 273
6.5.3 PCR and PLS Regression . . . 280
6.6 Exercises . . . 283

7 Moving Beyond Linearity 289
7.1 Polynomial Regression . . . 290
7.2 Step Functions . . . 292
7.3 Basis Functions . . . 293
7.4 Regression Splines . . . 294
7.4.1 Piecewise Polynomials . . . 294
7.4.2 Constraints and Splines . . . 296
7.4.3 The Spline Basis Representation . . . 296
7.4.4 Choosing the Number and Locations of the Knots . . . 297
7.4.5 Comparison to Polynomial Regression . . . 299

7.5 Smoothing Splines 300
7.5.1 An Overview of Smoothing Splines 300
7.5.2 Choosing the Smoothing Parameter λ 301
7.6 Local Regression 303
7.7 Generalized Additive Models 305
7.7.1 GAMs for Regression Problems 306
7.7.2 GAMs for Classification Problems 308
7.8 Lab: Non-Linear Modeling 309
7.8.1 Polynomial Regression and Step Functions 310
7.8.2 Splines 315
7.8.3 Smoothing Splines and GAMs 317
7.8.4 Local Regression 324
7.9 Exercises 325

8 Tree-Based Methods 331
8.1 The Basics of Decision Trees 331
8.1.1 Regression Trees 331
8.1.2 Classification Trees 337
8.1.3 Trees Versus Linear Models 341
8.1.4 Advantages and Disadvantages of Trees 341
8.2 Bagging, Random Forests, Boosting, and Bayesian Additive Regression Trees 343
8.2.1 Bagging 343
8.2.2 Random Forests 346
8.2.3 Boosting 347
8.2.4 Bayesian Additive Regression Trees 350
8.2.5 Summary of Tree Ensemble Methods 353
8.3 Lab: Tree-Based Methods 354
8.3.1 Fitting Classification Trees 355
8.3.2 Fitting Regression Trees 358
8.3.3 Bagging and Random Forests 360
8.3.4 Boosting 361
8.3.5 Bayesian Additive Regression Trees 362
8.4 Exercises 363

9 Support Vector Machines 367
9.1 Maximal Margin Classifier 367
9.1.1 What Is a Hyperplane? 368
9.1.2 Classification Using a Separating Hyperplane . . . 368
9.1.3 The Maximal Margin Classifier 370
9.1.4 Construction of the Maximal Margin Classifier . . 372
9.1.5 The Non-separable Case 372
9.2 Support Vector Classifiers 373
9.2.1 Overview of the Support Vector Classifier 373
9.2.2 Details of the Support Vector Classifier 374
9.3 Support Vector Machines 377
9.3.1 Classification with Non-Linear Decision Boundaries 378
9.3.2 The Support Vector Machine 379

9.3.3 An Application to the Heart Disease Data 382
9.4 SVMs with More than Two Classes 383
9.4.1 One-Versus-One Classification 384
9.4.2 One-Versus-All Classification 384
9.5 Relationship to Logistic Regression 384
9.6 Lab: Support Vector Machines 387
9.6.1 Support Vector Classifier 387
9.6.2 Support Vector Machine 390
9.6.3 ROC Curves 392
9.6.4 SVM with Multiple Classes 393
9.6.5 Application to Gene Expression Data 394
9.7 Exercises 395

10 Deep Learning 399
10.1 Single Layer Neural Networks 400
10.2 Multilayer Neural Networks 402
10.3 Convolutional Neural Networks 406
10.3.1 Convolution Layers 407
10.3.2 Pooling Layers 410
10.3.3 Architecture of a Convolutional Neural Network . . 410
10.3.4 Data Augmentation 411
10.3.5 Results Using a Pretrained Classifier 412
10.4 Document Classification 413
10.5 Recurrent Neural Networks 416
10.5.1 Sequential Models for Document Classification . . 418
10.5.2 Time Series Forecasting 420
10.5.3 Summary of RNNs 424
10.6 When to Use Deep Learning 425
10.7 Fitting a Neural Network 427
10.7.1 Backpropagation 428
10.7.2 Regularization and Stochastic Gradient Descent . . 429
10.7.3 Dropout Learning 431
10.7.4 Network Tuning 431
10.8 Interpolation and Double Descent 432
10.9 Lab: Deep Learning 435
10.9.1 Single Layer Network on Hitters Data 437
10.9.2 Multilayer Network on the MNIST Digit Data . . . 444
10.9.3 Convolutional Neural Networks 448
10.9.4 Using Pretrained CNN Models 452
10.9.5 IMDB Document Classification 454
10.9.6 Recurrent Neural Networks 458
10.10 Exercises 465

11 Survival Analysis and Censored Data 469
11.1 Survival and Censoring Times 470
11.2 A Closer Look at Censoring 470
11.3 The Kaplan–Meier Survival Curve 472
11.4 The Log-Rank Test 474
11.5 Regression Models With a Survival Response 476

11.5.1 The Hazard Function 476
11.5.2 Proportional Hazards 478
11.5.3 Example: Brain Cancer Data 482
11.5.4 Example: Publication Data 482
11.6 Shrinkage for the Cox Model 484
11.7 Additional Topics 486
11.7.1 Area Under the Curve for Survival Analysis 486
11.7.2 Choice of Time Scale 487
11.7.3 Time-Dependent Covariates 488
11.7.4 Checking the Proportional Hazards Assumption . . 488
11.7.5 Survival Trees 488
11.8 Lab: Survival Analysis 489
11.8.1 Brain Cancer Data 489
11.8.2 Publication Data 493
11.8.3 Call Center Data 494
11.9 Exercises 498

12 Unsupervised Learning 503
12.1 The Challenge of Unsupervised Learning 503
12.2 Principal Components Analysis 504
12.2.1 What Are Principal Components? 505
12.2.2 Another Interpretation of Principal Components . 508
12.2.3 The Proportion of Variance Explained 510
12.2.4 More on PCA 512
12.2.5 Other Uses for Principal Components 515
12.3 Missing Values and Matrix Completion 515
12.4 Clustering Methods 520
12.4.1 K-Means Clustering 521
12.4.2 Hierarchical Clustering 525
12.4.3 Practical Issues in Clustering 532
12.5 Lab: Unsupervised Learning 535
12.5.1 Principal Components Analysis 535
12.5.2 Matrix Completion 539
12.5.3 Clustering 542
12.5.4 NCI60 Data Example 546
12.6 Exercises 552

13 Multiple Testing 557
13.1 A Quick Review of Hypothesis Testing 558
13.1.1 Testing a Hypothesis 558
13.1.2 Type I and Type II Errors 562
13.2 The Challenge of Multiple Testing 563
13.3 The Family-Wise Error Rate 565
13.3.1 What is the Family-Wise Error Rate? 565
13.3.2 Approaches to Control the Family-Wise Error Rate 567
13.3.3 Trade-Off Between the FWER and Power 572
13.4 The False Discovery Rate 573
13.4.1 Intuition for the False Discovery Rate 573
13.4.2 The Benjamini–Hochberg Procedure 575

13.5 A Re-Sampling Approach to p-Values and False Discovery Rates . . . 577
13.5.1 A Re-Sampling Approach to the p-Value . . . 578
13.5.2 A Re-Sampling Approach to the False Discovery Rate 579
13.5.3 When Are Re-Sampling Approaches Useful? . . . 581
13.6 Lab: Multiple Testing . . . 583
13.6.1 Review of Hypothesis Tests . . . 583
13.6.2 Family-Wise Error Rate . . . 585
13.6.3 False Discovery Rate . . . 588
13.6.4 A Re-Sampling Approach . . . 590
13.7 Exercises . . . 593

Index **597**